



Target SQL Business Case (Google BigQuery and Looker Studio)



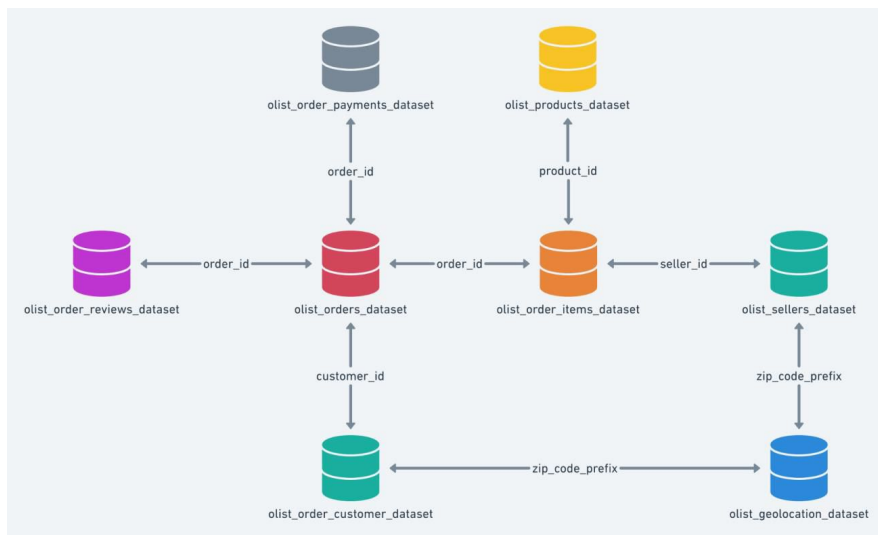
BigQuery



Why this case study?

Target is a globally renowned brand and a prominent retailer in the United States.

This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders. It consists of 8 csv files analysed in Google BigQuery and Looker Studio.



Dataset schema

Initial data exploration

--Trend in the no. of orders placed over the past years.

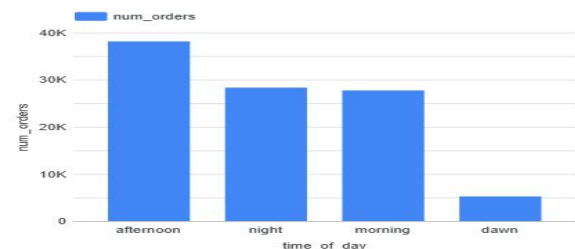
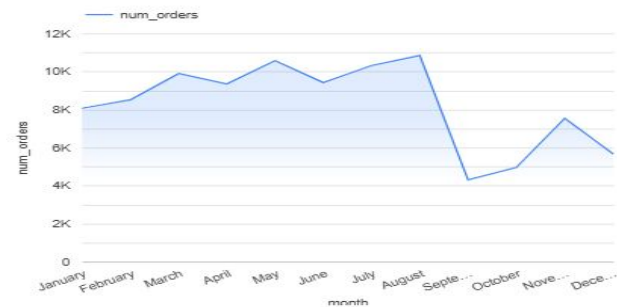
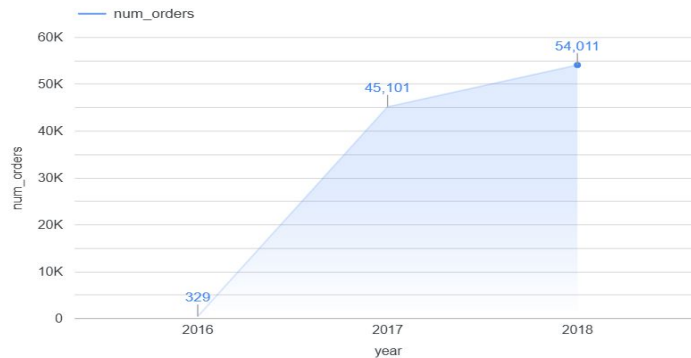
```
SELECT EXTRACT(year from order_purchase_timestamp) AS year,  
       EXTRACT(month from order_purchase_timestamp) AS month,  
       count(*) AS num_orders  
FROM `target_sql.orders`  
GROUP BY year, month  
ORDER BY year, month;
```

--Trend in the no. of orders placed over the months.

```
SELECT EXTRACT(month from order_purchase_timestamp) AS month,  
       count(*) AS num_orders  
FROM `target_sql.orders`  
GROUP BY month
```

--Trend by the hour of the day (Dawn, Morning, Afternoon, Night)

```
SELECT  
  CASE WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 0 and 6 THEN  
    'dawn'  
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 7 and 12 THEN  
    'morning'  
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 13 and 18 THEN  
    'afternoon'  
    ELSE 'night'  
  END AS time_of_day,  
  Count(*) as num_orders  
FROM `target_sql.orders`  
GROUP BY time_of_day  
ORDER BY num_orders DESC;
```



Evolution of E-commerce orders in Brazil region

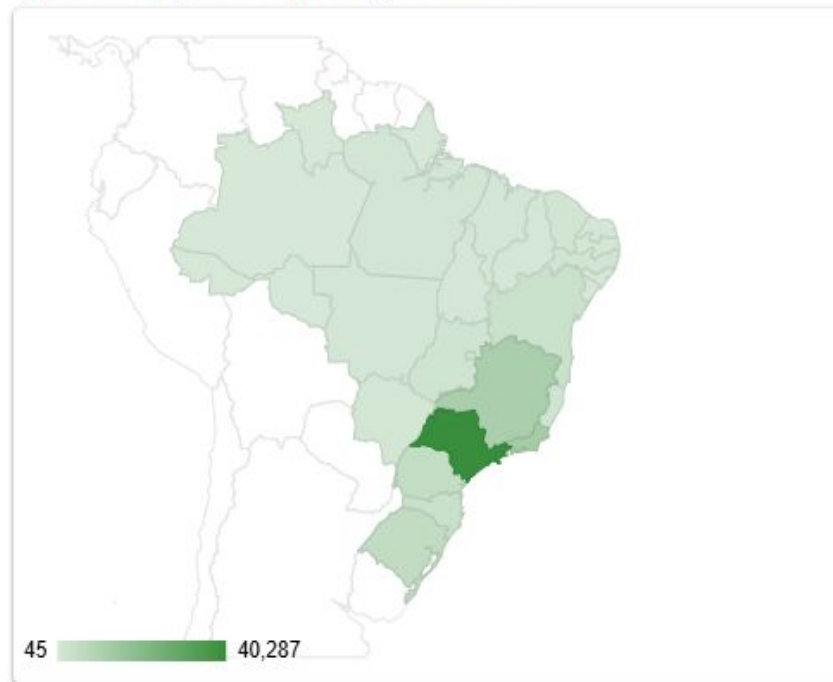
--Get month on month orders by states.

```
SELECT c.customer_state,  
       EXTRACT(month from order_purchase_timestamp) as month,  
       COUNT(*) AS num_orders  
FROM `target_sql.orders` o  
JOIN `target_sql.customers` c  
on o.customer_id=c.customer_id  
GROUP BY c.customer_state, month  
ORDER BY num_orders DESC;
```

--Distribution of customers across the states in Brazil

```
SELECT g.geolocation_state,  
       COUNT(DISTINCT(c.customer_unique_id)) AS num_customers  
FROM `target_sql.customers` c  
JOIN `target_sql.geolocation` g  
ON c.customer_zip_code_prefix=g.geolocation_zip_code_prefix  
GROUP BY g.geolocation_state  
ORDER BY num_customers DESC;
```

geolocation_state by num_customers



Majority of customers are located in Sao Paulo, Rio De Janeiro and Minas Gerais while lowest number of customers in Roraima, Amapá, Acre

Impact on Economy

```
--% Increase in cost from 2017 to 2018
WITH base AS(
  SELECT EXTRACT(year from o.order_purchase_timestamp) AS year,
         EXTRACT(month from o.order_purchase_timestamp) AS month,
         SUM(p.payment_value) as cost,
  FROM `target_sql.orders` o
  JOIN `target_sql.payments` p
  ON o.order_id=p.order_id
  WHERE EXTRACT(year from o.order_purchase_timestamp) BETWEEN 2017 AND 2018
        AND EXTRACT(month from o.order_purchase_timestamp) BETWEEN 1 AND 8
  GROUP BY year,month
  ORDER BY year,month
),
base_2 AS(
  SELECT year,
         SUM(cost) as cost
  FROM base
  GROUP BY year
  ORDER BY year
),
base_3 AS(
  SELECT * , LAG(cost,1) OVER(ORDER BY year) as last_year_cost
  FROM base_2
)
SELECT *,(cost-coalesce(last_year_cost,0))/coalesce(last_year_cost,1) *100 as per_inc
FROM base_3
WHERE year= 2018;
```

Row	year	cost	last_year_cost	per_inc
1	2018	8694733.840000...	3669022.119999...	136.98

--Total & Average value of order price,order freight for each state

```
WITH CTE AS
  (SELECT oi.order_id,oi.price, oi.freight_value,
   o.customer_id, c.customer_state
  FROM `target_sql.order_items` oi
  JOIN `target_sql.orders` o
  ON oi.order_id=o.order_id
  JOIN `target_sql.customers` c
  ON o.customer_id = c.customer_id)
```

```
SELECT customer_state,
sum(price) as
sum_price,sum(price)/count(distinct(order_id)) as
mean_price,
sum(freight_value) as sum_freight_value
,sum(freight_value)/count(distinct(order_id)) as
mean_freight_value,
FROM CTE
GROUP BY customer_state;
```

Row	customer_state	sum_price	mean_price	sum_freight_value	mean_freight_value
1	SP	5202955.049999...	125.7511794561...	718723.0700000...	17.37095033232...
2	RJ	1824092.669999...	142.9315679360...	305589.3100000...	23.94525231155...
3	PR	683083.760000001	136.6714205682...	117851.6799999...	23.57976790716...
4	RS	750304.0200000...	138.1266605301...	135522.7399999...	24.94895802650...
5	MG	1585308.030000...	137.3274454261...	270853.4600000...	23.46270443520...
6	GO	294591.9499999...	146.7822371699...	53114.98000000...	26.46486297957...

Analysis based on sales, freight and delivery time

```
--Days between  
purchasing,estimated,actual delivery date  
select order_id,  
TIMESTAMP_DIFF(order_delivered_customer_d  
ate,order_purchase_timestamp, DAY) AS  
time_to_delivery,  
TIMESTAMP_DIFF(order_delivered_customer_d  
ate,order_estimated_delivery_date, DAY)  
AS diff_estimated_delivery  
from `target_sql.orders`  
where order_status='delivered';
```

Row	order_id ▾	time_to_delivery ▾	diff_estimated_d...
1	bfb0f9bdef84302105ad712db...	54	36
2	98974b076b01553d49ee64679...	43	-6
3	c4b41c36dd589e901f6879f25a...	36	-14
4	d2292ff2201e74c5db154d1b7a...	29	-20
5	95e01270fcbae986342340010...	30	-19

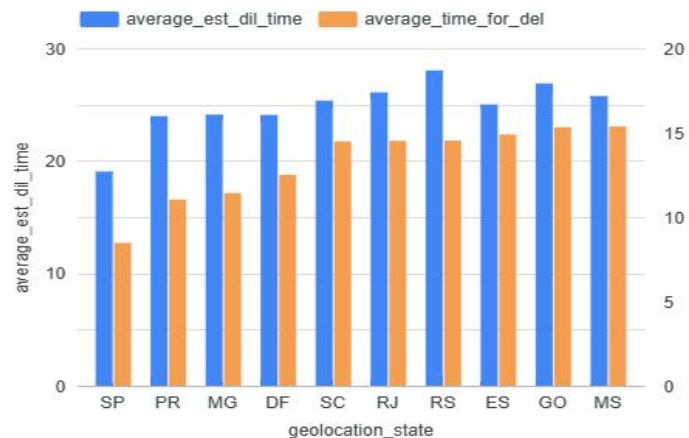
--Top 5 states with highest/lowest average time to delivery

```
select g.geolocation_state as state,  
SUM(TIMESTAMP_DIFF(  
order_delivered_customer_date,order_purchase_ti  
mestamp,  
DAY))/COUNT(ORDER_ID) AS avg_dil_time,  
from `target_sql.orders` o  
inner join `target_sql.customers` c  
on o.customer_id=c.customer_id  
inner join `target_sql.geolocation` g  
on  
c.customer_zip_code_prefix=g.geolocation_zip_co  
de_prefix  
where order_status='delivered'  
group by state  
order by avg_dil_time  
limit 5
```

Row	state ▾	avg_dil_time ▾
1	SP	8.468892914352...
2	PR	11.03876404770...
3	MG	11.41821678372...
4	DF	12.49651789233...
5	SC	14.48408434580...

```
--Top 5 states where delivery is really fast/ not so
fast compared to estimated date
select geolocation_state,
SUM(TIMESTAMP_DIFF(order_delivered_customer_date,order
_purchase_timestamp,
DAY))/COUNT(ORDER_ID) AS average_time_for_del,
SUM(TIMESTAMP_DIFF(order_estimated_delivery_date,
order_purchase_timestamp,
DAY))/COUNT(ORDER_ID) AS average_est_dil_time,
from `target_sql.orders` o
inner join `target_sql.customers` c
on o.customer_id=c.customer_id
inner join `target_sql.geolocation` g
on
c.customer_zip_code_prefix=g.geolocation_zip_code_pref
ix
where order_status='delivered'
group by geolocation_state
order by (average_time_for_del-average_est_dil_time)
```

	geolocation...	average_est_d... ^	average_time_fo...
1.	SP	19.05	8.47
2.	PR	23.97	11.04
3.	DF	24.08	12.5
4.	MG	24.11	11.42
5.	ES	25.01	14.89
6.	SC	25.35	14.48
7.	MS	25.75	15.37
8.	RJ	26.08	14.52
9.	GO	26.88	15.31
10.	TO	27.9	16.3
11.	RS	28.04	14.53



Analysis based on the payments:

--Orders for different payment types

Month over Month

```
SELECT payment_type, COUNT(o.order_id) as  
order_count,  
Extract( month from  
order_purchase_timestamp) as month,  
Extract( year from  
order_purchase_timestamp) as year,  
FROM `target_sql.payments` p  
join `target_sql.orders` o  
on o.order_id=p.order_id  
GROUP BY payment_type, year, month  
order by year, month;
```

Row	payment_type ▾	order_count ▾	month ▾	year ▾
1	credit_card	3	9	2016
2	voucher	23	10	2016
3	debit_card	2	10	2016
4	UPI	63	10	2016
5	credit_card	254	10	2016
6	credit_card	1	12	2016
7	UPI	197	1	2017
8	debit_card	9	1	2017

--Count of orders based on the no. of
payment installments

```
SELECT distinct(payment_installments) as  
installments, count(order_id) as  
Num_orders,  
FROM `target_sql.payments`  
where payment_installments>1  
GROUP BY payment_installments  
order by Num_orders desc;
```

Row	installments ▾	Num_orders ▾
1	2	12413
2	3	10461
3	4	7098
4	10	5328
5	5	5239
6	8	4268
7	6	3920
8	7	1626
9	9	644

Insights and recommendations:

Insights -

- 1) Sales are highest in November
- 2) Sales are highest from 12-6pm
- 3) Majority of customers are located in Sao Paulo, Rio De Janeiro and Minas Gerais while lowest number of customers in Roraima, Amapá, Acre
- 4) The average time to delivery (between purchase and delivery) is about 12 days.
- 5) The average time between estimated and actual delivery is about 10 days i.e. products arrive 10 days earlier than expected on an average.
- 6) Roraima has highest mean of freight value.
- 7) São Paulo has lowest mean of freight value.
- 8) Roraima has highest mean time to delivery.
- 9) São Paulo has lowest mean time to delivery.
- 10) Maximum number of orders were placed with just a single payment installment.
- 11) There is a 137% increase in cost of orders from 2017 to 2018 (for Jan-Aug months).

Recommendations-

- 1) Stock up inventory in November
- 2) Clearance sale in November
- 3) Offer more deals during afternoon peak hours
- 4) More deals, varieties and offers in states having majority customers like Sao Paulo, Rio De Janeiro and Minas Gerais
- 5) More deals and promotions in states like Sao Paulo where time to delivery is lowest.